Deepjyoti Deka

Research Scientist

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Summary

Dr. Deka is a research scientist and project manager working in Machine Learning for smart grids and cyber-physical networks. His research leverages network physics and statistics to design certifiable algorithms for network identification, grid optimization and de-carbonization.
Citizenship: India, **Permanent Residence:** USA

Education

2011 – 2015 **Ph.D. in Electrical & Computer Engineering**, *The University of Texas at Austin, Austin, TX.*
Dissertation: Analysis of the power grid: structure & secure operations.
Adviser: Prof. Sriram Vishwanath, (Communication and Networks Track).

2009 – 2011 **M.S. in Electrical & Computer Engineering**, *The University of Texas at Austin, Austin, TX.*
CGPA: $3.94/4$.

2005 – 2009 **B.Tech. in Electronics & Comm. Engineering**, *Indian Institute of Technology (IIT) Guwahati, India.* CGPA: 9.58/10, **Institute Silver medal** for highest graduating CGPA in department.
Thesis: Power Control and Lifetime Maximization Techniques For Cooperative Communication.

Research Interests

Machine Learning: Graphical models, Deep learning, Time-series, Data-driven Optimization

Energy Systems: Renewable integration, Datacenter planning, Load Flexibility & Cyber-Security.

Awards and Achievements

2022 **Young Alumni Achiever Award,** IIT Guwahati, India.

2021 **Best Paper on ML Innovation** at ICML Workshop on Climate Change AI.

2015 **Best Conference Papers on Cyber Security, Stability, and Protection** at IEEE PES GM.

2015 **Student Leadership Award** by School of Engineering, University of Texas Austin.

2011 IEEE Communication Society **Student Travel Grant**.

2009 **Institute Silver Medal** for best graduating student in Electronics and Communication Engineering at IIT Guwahati.

2008 – 2009 **OP Jindal Engineering and Management scholarship** awarded to 25 students across India.

2006 – 2008 **Institute Merit Scholarship** in academic years $2006-07$ and $2007-08$ for securing first rank in the department at IIT Guwahati.

2003 **NTSE merit scholarship** for higher education, awarded by the Government of India.

2003 **Gold Medal,** Regional Mathematics Olympiad in Assam, India.

Professional Experience

July 2024 – present **Research Scientist and Program Manager**, *MIT Energy Initiative, MIT*, Cambridge, Massachusetts,
PI/co-PI of projects on new techniques for operations and planning for datacenter growth in renewable-rich power grids.
Mentoring: 1 postdoc, 3 summer students
Technical Projects: (a) Machine Learning for feasible grid operations under topology changes,
(b) Co-optimizing datacenter flexibility and storage for market operations and reliability,
(c) Agentic AI tools for surrogate model development for use in network optimization.

July 2022 – July 2024 **Scientist 3**, *Theoretical (T) Division, Los Alamos National Laboratory*, Los Alamos, New Mexico,
Project: PI/co-PI of multiple DOE and LANL projects on applications of machine learning in cyber-physical systems and infrastructure network optimization.
Mentoring: 3 postdocs, 25+ summer students
Technical Projects: (a) Machine Learning augmented optimization for power systems,
(b) Robust neural-networks for network change detection and fault localization,
(c) Control of interdependent infrastructure networks and natural systems for reliability.

September 2018-June 2022 **Scientist 2**, *Theoretical (T) Division, Los Alamos National Laboratory*, Los Alamos, New Mexico,
Technical Projects: (a) Physics-informed machine learning for Topology and Parameter estimation,
(b) Aggregated Control of flexible loads in power grids,
(c) Learning for Cyber-Physical Systems.

Dec. 2015 – Sept. 2018 **Postdoctoral Researcher**, *Center for Non-linear Studies (CNLS), Los Alamos National Laboratory*, Los Alamos, New Mexico. **Mentor:** Michael Chertkov, Theoretical (T) division.

May– Aug. 2015, 2014 **Summer Intern**, *Theory Division, Los Alamos National Laboratory*, Los Alamos, New Mexico,
Project: Non-intrusive algorithm design to learn topology and consumption profiles in power grids.

May 2013 – August 2013 **Summer Intern**, *Electric Reliability Council of Texas (ERCOT)*, Taylor, Texas,
Project: Storage incorporation into ERCOT through the Fast Responding Regulation Service Pilot.

May 2008 – July 2008 **Summer Intern**, *Corporate R & D, Qualcomm Inc.*, San Diego, CA,
Project: Modeling and improvement in page download times over wireless networks.

May 2007 – July 2007 **Student Intern**, *Indian Institute of Sciences*, Bangalore, India,
Project: Analysis of optimum space-time power allocation policy for MIMO communication systems.

Preprints

[1] Aron Brenner, Deepjyoti **Deka**, Line Roald, and Saurabh Amin. “Strategic Spatial Load Shifting and Market Efficiency”. 2026.

[2] Ali Jalilian, Deepjyoti **Deka**, Md Umar Hashmi, and Dirk Van Hertem. “Coordinated Dynamic Operating Envelopes for Unlocking Additional Flexibility at Grid Edge”. 2026.

[3] Shaohui Liu, Sungho Shin, and Deepjyoti **Deka**. “Watts vs. Bytes: Turning Data Centers into Grid Assets via Storage Compute Co-Optimization”. 2026.

[4] Abhay Singh Bhadoriya, Deepjyoti **Deka**, and Kaarthik Sundar. “A Family of Convex Models to Achieve Fairness through Dispersion Control”. 2025.

[5] Young-ho Cho, Harsha Nagarajan, Deepjyoti **Deka**, and Hao Zhu. “Sparse Neural Approximations for Bilevel Adversarial Problems in Power Grids”. 2025.

[6] Dirk Lauinger, Deepjyoti **Deka**, and Sungho Shin. “The value of storage in electricity distribution: The role of markets”. 2025.

[7] Abhay Singh Bhadoriya, Deepjyoti **Deka**, and Kaarthik Sundar. “Equitable Routing–Rethinking the Multiple Traveling Salesman Problem”. 2024.

Journal Articles

[8] Vincenzo Di Vito, Kaarthik Sundar, Ferdinando Fioretto, and Deepjyoti **Deka**. “Stability-constrained AC optimal power flow–A Gaussian process based approach”. In: *Sustainable Energy, Grids and Networks* 46 (2026).

[9] Jie Feng, Yuanyuan Shi, and Deepjyoti **Deka**. “Efficient Policy Adaptation for Voltage Control Under Unknown Topology Changes”. In: *Electric Power Systems Research* (2026). Special issue on Power Systems Computation Conference (PSCC), accepted.

[10] Aparna Kishore, Kaarthik Sundar, Deepjyoti **Deka**, and Madhav Marathe. “Tunable and computationally efficient framework for retrofitting: Budget allocation using income and spatial equity”. In: *PNAS Nexus* (May 2026).

[11] Parikshit Pareek, Sidhant Misra, and Deepjyoti **Deka**. “Learning Power Flow with Confidence: A Probabilistic Guarantee Framework for Voltage Risk”. In: *Electric Power Systems Research* (2026). Special issue on Power Systems Computation Conference (PSCC), accepted.

[12] Bendong Tan, Tong Su, Yu Weng, Ketian Ye, Parikshit Pareek, Petr Vorobev, Hung Nguyen, Junbo Zhao, and Deepjyoti **Deka**. "Gaussian processes in power systems: Techniques, applications, and future works". In: *Applied Energy* 402 (2026).

[13] Rohit Kannan, Harsha Nagarajan, and Deepjyoti **Deka**. "Strong Partitioning and a Machine Learning Approximation for Accelerating the Global Optimization of Nonconvex Quadratically Constrained Quadratic Programs". In: *INFORMS Journal on Computing* (2025).

[14] Parikshit Pareek, Sidhant Misra, and Deepjyoti **Deka**. "Data-efficient strategies for probabilistic voltage envelopes under network contingencies". In: *Sustainable Energy, Grids and Networks (IREP)* (2025), p. 101811.

[15] Anirudh Rayas, Jiajun Cheng, Rajasekhar Anguluri, Deepjyoti **Deka**, and Gautam Dasarathy. "Learning Networks From Wide-Sense Stationary Stochastic Processes". In: *IEEE Transactions on Signal and Information Processing over Networks* (2025).

[16] Kaarthik Sundar, Deepjyoti **Deka**, and Russell Bent. "A Parametric, second-order cone representable model of fairness for decision-making problems: K. Sundar et al." In: *Optimization and Engineering* 26 (2025).

[17] Bendong Tan, Deepjyoti **Deka**, and Junbo Zhao. "Spatial-Temporal Global Sensitivity Analysis for Probabilistic Transient Stability Assessment". In: *IEEE Transactions on Power Systems* (2025).

[18] Wenting Li, Deepjyoti **Deka**, Ren Wang, and Mario R Arrieta Paternina. "Physics-constrained adversarial training for neural networks in stochastic power grids". In: *IEEE Transactions on Artificial Intelligence* 5.3 (2023), pp. 1121–1131.

[19] Aleksander Lukashevich, Vyacheslav Gorchakov, Petr Vorobev, Deepjyoti **Deka**, and Yury Maximov. "Importance sampling approach to chance-constrained dc optimal power flow". In: *IEEE Transactions on Control of Network Systems* (2023).

[20] Mile Mitrovic, Ognjen Kundacina, Aleksandr Lukashevich, Semen Budennyy, Petr Vorobev, Vladimir Terzija, Yury Maximov, and Deepjyoti **Deka**. "Gp cc-opf: Gaussian process based optimization tool for chance-constrained optimal power flow". In: *Software Impacts* 16 (2023), p. 100489.

[21] Mile Mitrovic, Aleksandr Lukashevich, Petr Vorobev, Vladimir Terzija, Semen Budennyy, Yury Maximov, and Deepjyoti **Deka**. "Data-driven stochastic AC-OPF using Gaussian process regression". In: *International Journal of Electrical Power Energy Systems* 152 (2023), p. 109249.

[22] Venkat Ram Subramanian, Deepjyoti **Deka**, Saurav Talukdar, Andrew Lamperski, and Murti Salapaka. "Topology Learning in Radial Dynamical Systems With Unreliable Data". In: *IEEE Transactions on Control of Network Systems* 10.4 (2023), pp. 2010–2021.

[23] Deepjyoti **Deka**, Vassilis Kekatos, and Guido Cavraro. "Learning Distribution Grid Topologies: A Tutorial". In: *IEEE Transactions on Smart Grid* (2022).

[24] Sarthak Gupta, Sidhant Misra, Deepjyoti **Deka**, and Vassilis Kekatos. "DNN-based policies for stochastic AC OPF". In: *Electric Power Systems Research, Special issue on Power Systems Computation Conference (PSCC)* 213 (2022), p. 108563.

[25] Md Umar Hashmi, Deepjyoti **Deka**, Ana Busic, and Dirk Van Hertem. "Can locational disparity of prosumer energy optimization due to inverter rules be limited?" In: *IEEE Transactions on Power Systems* (2022).

[26] Guanze Peng, Robert Mieth, Deepjyoti **Deka**, and Yury Dvorkin. "Markovian Decentralized Ensemble Control for Demand Response". In: *IEEE Control Systems Letters* 6 (2022), pp. 3050–3055.

[27] Mohini Bariya, Deepjyoti **Deka**, and Alexandra von Meier. "Guaranteed phase & topology identification in three phase distribution grids". In: *IEEE Transactions on Smart Grid* 12.4 (2021), pp. 3605–3612.

[28] Siddharth Bhela, Harsha Nagarajan, Deepjyoti **Deka**, and Vassilis Kekatos. "Efficient topology design algorithms for power grid stability". In: *IEEE Control Systems Letters* 6 (2021), pp. 1100–1105.

[29] Deepjyoti **Deka**, Harish Doddi, Sidhant Misra, and Murti V Salapaka. "Tractable Learning in Underexcited Power Grids". In: *IEEE transactions on control of network systems* 9.2 (2021), pp. 763–774.

[30] Ali Hassan, Deepjyoti **Deka**, and Yury Dvorkin. “Privacy-aware load ensemble control: A linearly-solvable MDP approach”. In: *IEEE Transactions on Smart Grid* 13.1 (2021), pp. 255–267.

[31] Ignacio Losada Carreno, Anna Scaglione, Anatoly Zlotnik, Deepjyoti **Deka**, and Kaarthik Sundar. “An adversarial model for attack vector vulnerability analysis on power and gas delivery operations”. In: *Electric Power Systems Research, Special issue on Power Systems Computation Conference (PSCC)* 189 (2020), p. 106777.

[32] Deepjyoti **Deka**, Michael Chertkov, and Scott Backhaus. “Joint Estimation of Topology and Injection Statistics in Distribution Grids with Missing Nodes”. In: *IEEE Transactions on Control of Network Systems* (2020).

[33] Deepjyoti **Deka**, Saurav Talukdar, Michael Chertkov, and Murti V Salapaka. “Graphical models in meshed distribution grids: Topology estimation, change detection & limitations”. In: *IEEE Transactions on Smart Grid* 11.5 (2020), pp. 4299–4310.

[34] Md Umar Hashmi, Deepjyoti **Deka**, Ana Busic, Lucas Pereira, and Scott Backhaus. “Arbitrage with power factor correction using energy storage”. In: *IEEE Transactions on Power Systems* 35.4 (2020), pp. 2693–2703.

[35] Ali Hassan, Samrat Acharya, Michael Chertkov, Deepjyoti **Deka**, and Yury Dvorkin. “A hierarchical approach to multienergy demand response: From electricity to multienergy applications”. In: *Proceedings of the IEEE* 108.9 (2020), pp. 1457–1474.

[36] Ali Hassan, Deepjyoti **Deka**, Michael Chertkov, and Yury Dvorkin. “Data-driven learning and load ensemble control”. In: *Electric Power Systems Research, Special issue on Power Systems Computation Conference (PSCC)* 189 (2020), p. 106780.

[37] Ali Hassan, Robert Mieth, Deepjyoti **Deka**, and Yury Dvorkin. “Stochastic and distributionally robust load ensemble control”. In: *IEEE Transactions on Power Systems* 35.6 (2020), pp. 4678–4688.

[38] Ilyes Mezghani, Sidhant Misra, and Deepjyoti **Deka**. “Stochastic AC optimal power flow: A data-driven approach”. In: *Electric Power Systems Research, Special issue on Power Systems Computation Conference (PSCC)* 189 (2020), p. 106567.

[39] Sejun Park, Deepjyoti **Deka**, Scott Backhaus, and Michael Chertkov. “Learning with end-users in distribution grids: Topology and parameter estimation”. In: *IEEE Transactions on Control of Network Systems* 7.3 (2020), pp. 1428–1440.

[40] Nikolay Stulov, Dejan J Sobajic, Yury Maximov, Deepjyoti **Deka**, and Michael Chertkov. “Learning model of generator from terminal data”. In: *Electric Power Systems Research, Special issue on Power Systems Computation Conference (PSCC)* 189 (2020), p. 106742.

[41] Saurav Talukdar, Deepjyoti **Deka**, Harish Doddi, Donatello Materassi, Michael Chertkov, and Murti V Salapaka. “Physics informed topology learning in networks of linear dynamical systems”. In: *Automatica* 112 (2020), p. 108705.

[42] Deepjyoti **Deka**, Michael Chertkov, and Scott Backhaus. “Topology estimation using graphical models in multi-phase power distribution grids”. In: *IEEE Transactions on Power Systems* 35.3 (2019), pp. 1663–1673.

[43] Mana Jalali, Vassilis Kekatos, Nikolaos Gatsis, and Deepjyoti **Deka**. “Designing reactive power control rules for smart inverters using support vector machines”. In: *IEEE Transactions on Smart Grid* 11.2 (2019), pp. 1759–1770.

[44] Wenting Li, Deepjyoti **Deka**, Michael Chertkov, and Meng Wang. “Real-time Faulted Line Localization and PMU Placement in Power Systems through Convolutional Neural Networks”. In: *IEEE Transactions on Power Systems* 34.6 (2019), pp. 4640–4651.

[45] Deepjyoti **Deka**, Sriram Vishwanath, and Ross Baldick. “Topological vulnerability of power grids to disasters: Bounds, adversarial attacks and reinforcement”. In: *Plos one* 13.10 (2018), e0204815.

[46] Colin Grudzien, Deepjyoti **Deka**, Michael Chertkov, and Scott N Backhaus. “Structure-and physics-preserving reductions of power grid models”. In: *Multiscale Modeling & Simulation* 16.4 (2018), pp. 1916–1947.

[47] Ali Hassan, Robert Mieth, Michael Chertkov, Deepjyoti **Deka**, and Yury Dvorkin. "Optimal load ensemble control in chance-constrained optimal power flow". In: *IEEE Transactions on Smart Grid* 10.5 (2018), pp. 5186–5195.

[48] Deepjyoti **Deka**, Scott Backhaus, and Michael Chertkov. "Structure Learning in Power Distribution Networks". In: *IEEE Transactions on Control of Network Systems* (2017).

[49] Deepjyoti **Deka**, Sriram Vishwanath, and Ross Baldick. "Analytical Models for Power Networks: The case of the Western US and ERCOT grids". In: *IEEE Transactions on Smart Grid* 8.6 (2017), pp. 2794–2802.

Conference Papers (peer reviewed)

[50] Daniel Glover, Parikshit Pareek, Deepjyoti **Deka**, and Anamika Dubey. "Power flow approximations for multiphase distribution networks using Gaussian processes". In: *2025 IEEE Power & Energy Society General Meeting (PESGM)*. IEEE. 2025, pp. 1–5.

[51] Md Umar Hashmi, Brida V Mbuwir, Hussain Kazmi, Carlo Manna, and Deepjyoti **Deka**. "Aggregation and disaggregation of distributed flexibility for transmission network needs: a conceptual framework". In: *IET Conference Proceedings CP922*. Vol. 2025. 14. IET. 2025, pp. 1743–1747.

[52] Parikshit Pareek, Abhijith Jayakumar, Kaarthik Sundar, Sidhant Misra, and Deepjyoti **Deka**. "Optimization Proxies using Limited Labeled Data and Training Time–A Semi-Supervised Bayesian Neural Network Approach". In: *International Conference on Machine Learning*. PMLR. 2025, pp. 47953–47970.

[53] Yize Chen, Deepjyoti **Deka**, and Yuanyuan Shi. "Contributions of Individual Generators to Nodal Carbon Emissions". In: *ACM e-Energy*. 2024.

[54] Mishfad Shaikh Veedu, Deepjyoti **Deka**, and Murti Salapaka. "Information Theoretically Optimal Sample Complexity of Learning Dynamical Directed Acyclic Graphs". In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2024, pp. 4636–4644.

[55] Yuqi Zhou, Kaarthik Sundar, Hao Zhu, and Deepjyoti **Deka**. "Mitigating the Impact of Uncertain Wildfire Risk on Power Grids through Topology Control". In: *International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*. IEEE. 2024, pp. 1–6.

[56] Mile Mitrovic, Aleksandr Lukashevich, Petr Vorobev, Vladimir Terzija, Yury Maximov, and Deepjyoti **Deka**. "Fast Data-Driven Chance Constrained AC-OPF Using Hybrid Sparse Gaussian Processes". In: *IEEE PowerTech, Belgrade*. IEEE. 2023, pp. 1–7.

[57] Harish Doddi, Deepjyoti **Deka**, Saurav Talukdar, and Murti Salapaka. "Efficient and passive learning of networked dynamical systems driven by non-white exogenous inputs". In: *International Conference on Artificial Intelligence and Statistics*. Vol. 151. PMLR. 2022, pp. 9982–9997.

[58] Mohasinina Kamal, Wenting Li, Deepjyoti **Deka**, and Hamed Mohsenian-Rad. "Physics-conditioned generative adversarial networks for state estimation in active power distribution systems with low observability". In: *2022 International Conference on Smart Grid Synchronized Measurements and Analytics (SGSMA)*. IEEE. 2022, pp. 1–6.

[59] Wenting Li and Deepjyoti **Deka**. "PPGN: Physics-Preserved Graph Networks for Real-Time Fault Location in Distribution Systems with Limited Observation and Labels". In: *Hawaii International Conference on System Sciences*. 2022.

[60] Ignacio Losada Carreno, Anna Scaglione, Anthony Giacomoni, Kaarthik Sundar, Deepjyoti **Deka**, and Anatoly Zlotnik. "Using Transient Pipeline Simulation to Evaluate Electric Power Generation Reliability". In: *PSIG Annual Meeting*. PSIG. 2021, PSIG–2119.

[61] Christopher Hannon, Deepjyoti **Deka**, Dong Jin, Marc Vuffray, and Andrey Y Lokhov. "Real-time anomaly detection and classification in streaming PMU data". In: *2021 IEEE Madrid PowerTech*. IEEE. 2021, pp. 1–6.

[62] Wenting Li and Deepjyoti **Deka**. "Physics-Informed Learning for High Impedance Faults Detection". In: *IEEE Powertech*. 2021.

[63] Sina Sontowski, Nigel Lawrence, Deepjyoti **Deka**, and Maanak Gupta. "Detecting anomalies using overlapping electrical measurements in smart power grids". In: *2021 IEEE International Conference on Big Data (Big Data)*. IEEE. 2021, pp. 2434–2441.

[64] Harish Doddi, Deepjyoti **Deka**, and Murti Salapaka. "Learning partially observed meshed distribution grids". In: *2020 International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*. IEEE. 2020, pp. 1–6.

[65] Md Umar Hashmi, Deepjyoti **Deka**, Lucas Pereira, and Ana Busic. "Energy Storage Optimization for Grid Reliability". In: *Proceedings of the Eleventh ACM International Conference on Future Energy Systems*. 2020, pp. 516–522.

[66] Siddharth Bhela, Deepjyoti **Deka**, Harsha Nagarajan, and Vassilis Kekatos. "Designing Power Grid Topologies for Minimizing Network Disturbances: An Exact MILP Formulation". In: *2019 American Control Conference (ACC)*. IEEE. 2019, pp. 1949–1956.

[67] Yize Chen, Md Umar Hashmi, Deepjyoti **Deka**, and Michael Chertkov. "Stochastic battery operations using deep neural networks". In: *IEEE ISGT NA 2019-Innovative Smart Grid Technologies Conference*. 2019.

[68] Deepjyoti **Deka** and Sidhant Misra. "Learning for DC-OPF: Classifying active sets using neural nets". In: *2019 IEEE Milan PowerTech*. IEEE. 2019, pp. 1–6.

[69] Harish Doddi, Saurav Talukdar, Deepjyoti **Deka**, and Murti Salapaka. "Exact topology learning in a network of cyclostationary processes". In: *2019 American Control Conference (ACC)*. IEEE. 2019, pp. 4968–4973.

[70] Md Umar Hashmi, Deepjyoti **Deka**, Ana Busic, Lucas Pereira, and Scott Backhaus. "Co-optimizing energy storage for prosumers using convex relaxations". In: *2019 International Conference on Intelligent System Application to Power Systems (ISAP)*. IEEE. 2019, pp. 1–7.

[71] Yize Chen, Yushi Tan, and Deepjyoti **Deka**. "Is machine learning in power systems vulnerable". In: *2018 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*. IEEE. 2018, pp. 1–6.

[72] Michael Chertkov, Deepjyoti **Deka**, and Yury Dvorkin. "Optimal ensemble control of loads in distribution grids with network constraints". In: *2018 Power Systems Computation Conference (PSCC)*. IEEE. 2018, pp. 1–7.

[73] Harish Doddi, Saurav Talukdar, Deepjyoti **Deka**, and Murti Salapaka. "Data-driven identification of a thermal network in multi-zone building". In: *2018 IEEE Conference on Decision and Control (CDC)*. IEEE. 2018, pp. 7302–7307.

[74] Ali Hassan, Yury Dvorkin, Deepjyoti **Deka**, and Michael Chertkov. "Chance-constrained ADMM approach for decentralized control of distributed energy resources". In: *2018 Power Systems Computation Conference (PSCC)*. IEEE. 2018, pp. 1–7.

[75] Andrey Y Lokhov, Deepjyoti **Deka**, Marc Vuffray, and Michael Chertkov. "Uncovering power transmission dynamic model from incomplete pmu observations". In: *2018 IEEE Conference on Decision and Control (CDC)*. IEEE. 2018, pp. 4008–4013.

[76] Andrey Y Lokhov, Marc Vuffray, Dmitry Shemetov, Deepjyoti **Deka**, and Michael Chertkov. "Online learning of power transmission dynamics". In: *2018 Power Systems Computation Conference (PSCC)*. IEEE. 2018, pp. 1–7.

[77] Seiun Park, Deepjyoti **Deka**, and Michael Chertkov. "Exact topology and parameter estimation in distribution grids with minimal observability". In: *2018 power systems computation conference (PSCC)*. IEEE. 2018, pp. 1–6.

[78] Sejun Park, Deepjyoti **Deka**, et al. "Learning in power distribution grids under correlated injections". In: *2018 52nd Asilomar Conference on Signals, Systems, and Computers*. IEEE. 2018, pp. 1863–1868.

[79] Saurav Talukdar, Deepjyoti **Deka**, Michael Chertkov, and Murti Salapaka. "Topology learning of radial dynamical systems with latent nodes". In: *2018 Annual American Control Conference (ACC)*. IEEE. 2018, pp. 1096–1101.

[80] D **Deka**, M Chertkov, S Talukdar, and MV Salapaka. "Topology estimation in bulk power grids: Theoretical guarantees and limits". In: *Bulk Power Systems Dynamics and Control Symposium-IREP*. 2017.

[81] Deepjyoti **Deka**, Michael Chertkov, and Scott Backhaus. "Estimating topology and injection statistics in distribution grids with hidden nodes". In: *2017 IEEE International Conference on Smart Grid Communications (SmartGridComm)*. IEEE. 2017, pp. 71–76.

[82] Deepjyoti **Deka**, Harsha Nagarajan, and Scott Backhaus. "Optimal topology design for disturbance minimization in power grids". In: *2017 American Control Conference (ACC)*. IEEE. 2017, pp. 2719–2724.

[83] Deepjyoti **Deka**, Armin Zare, Andrey Lokhov, Mihailo Jovanovic, and Michael Chertkov. "State and noise covariance estimation in power grids using limited nodal pmus". In: *2017 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*. IEEE. 2017, pp. 1075–1079.

[84] Emma M Stewart, Philip Top, Michael Chertkov, Deepjyoti **Deka**, Scott Backhaus, Andrey Lokhov, Ciaran Roberts, Val Hendrix, Sean Peisert, Anthony Florita, et al. "Integrated multi-scale data analytics and machine learning for the distribution grid". In: *2017 IEEE International Conference on Smart Grid Communications (SmartGridComm)*. IEEE. 2017, pp. 423–429.

[85] Saurav Talukdar, Deepjyoti **Deka**, Sandeep Attree, Donatello Materassi, and Murti Salapaka. "Learning the exact topology of undirected consensus networks". In: *2017 IEEE 56th Annual Conference on Decision and Control (CDC)*. IEEE. 2017, pp. 5784–5789.

[86] Saurav Talukdar, Deepjyoti **Deka**, Blake Lundstrom, Michael Chertkov, and Murti V Salapaka. "Learning exact topology of a loopy power grid from ambient dynamics". In: *Proceedings of the eighth international conference on future energy systems*. 2017, pp. 222–227.

[87] Saurav Talukdar, Deepjyoti **Deka**, Donatello Materassi, and Murti Salapaka. "Exact topology reconstruction of radial dynamical systems with applications to distribution system of the power grid". In: *2017 American Control Conference (ACC)*. IEEE. 2017, pp. 813–818.

[88] Deepjyoti **Deka**, Scott Backhaus, and Michael Chertkov. "Estimating distribution grid topologies: A graphical learning based approach". In: *2016 Power Systems Computation Conference (PSCC)*. IEEE. 2016, pp. 1–7.

[89] Deepjyoti **Deka**, Scott Backhaus, and Michael Chertkov. "Learning topology of distribution grids using only terminal node measurements". In: *2016 IEEE International Conference on Smart Grid Communications (SmartGridComm)*. IEEE. 2016, pp. 205–211.

[90] Deepjyoti **Deka**, Scott Backhaus, and Michael Chertkov. "Learning topology of the power distribution grid with and without missing data". In: *2016 European Control Conference (ECC)*. IEEE. 2016, pp. 313–320.

[91] Deepjyoti **Deka**, Scott Backhaus, and Michael Chertkov. "Tractable structure learning in radial physical flow networks". In: *2016 IEEE 55th Conference on Decision and Control (CDC)*. IEEE. 2016, pp. 6631–6638.

[92] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "Jamming aided generalized data attacks: Exposing vulnerabilities in secure estimation". In: *2016 49th Hawaii International Conference on System Sciences (HICSS)*. IEEE. 2016, pp. 2556–2565.

[93] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "Data attacks on power grids: Leveraging detection". In: *2015 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE. 2015, pp. 1–5.

[94] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "One breaker is enough: Hidden topology attacks on power grids". In: *2015 IEEE Power & Energy Society General Meeting*. IEEE. 2015, pp. 1–5.

[95] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "Optimal data attacks on power grids: Leveraging detection & measurement jamming". In: *2015 IEEE International Conference on Smart Grid Communications (SmartGridComm)*. IEEE. 2015, pp. 392–397.

[96] Deepjyoti **Deka** and Sriram Vishwanath. "Structural vulnerability of power grids to disasters: Bounds and reinforcement measures". In: *2015 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*. IEEE. 2015, pp. 1–5.

[97] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "Attacking power grids with secure meters: The case for breakers and jammers". In: *2014 IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. IEEE. 2014, pp. 646–651.

[98] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "Data attack on strategic buses in the power grid: Design and protection". In: *2014 IEEE PES General Meeting*. IEEE. 2014, pp. 1–5.

[99] Deepjyoti **Deka**, Ross Baldick, and Sriram Vishwanath. "Optimal hidden SCADA attacks on power grid: A graph theoretic approach". In: *2014 International conference on computing, networking and communications (ICNC)*. IEEE. 2014, pp. 36–40.

[100] Deepjyoti **Deka** and Sriram Vishwanath. "Generative growth model for power grids". In: *2013 International Conference on Signal-Image Technology & Internet-Based Systems*. IEEE. 2013, pp. 591–598.

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Book Chapters and Technical Reports

[102] Deepjyoti **Deka** and Michael Chertkov. "Topology detection in distribution networks with machine learning". In: *Big data application in power systems*. Elsevier Science, 2024, pp. 89–108.

[103] H Mohsenian-Rad, JY Joo, M Balestrieri, L Piyasinghe, S Biswas, K Chatterjee, H Valizadeh-Haghi, K Chang, J Grappe, T Laughner, et al. "Synchro-Waveform Measurements and Data Analytics in Power Systems". In: IEEE PES Technical Report. 2024.

[104] Reza Arghandeh, Mohini Bariya, George Cotter, Deepjyoti **Deka**, Sai Akhil Reddy Konakalla, Younes Seyedi, and Alireza Shasavari. "Synchronized measurements and their applications in distribution systems: An update". In: 2020.

[105] Reza Arghandeh, Mohini Bariya, George Cotter, Deepjyoti **Deka**, Sai Akhil Reddy Konakalla, Younes Seyedi, and Alireza Shasavari. "Synchrophasor Monitoring for Distribution Systems: Technical Foundations and Applications". In: 2018.

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[107] Emma Stewart, Karthik Chellappan, Scott Backhaus, Deepjyoti **Deka**, Matthew Reno, Sean Peisert, Dan Arnold, Chen Chen, Anthony Florita, and Mark Buckner. "Integrated Multi Scale Data Analytics and Machine Learning for the Grid: Benchmarking Algorithms and Data Quality Analysis". In: 2018.

[108] Emma M Stewart, Val Hendrix, Michael Chertkov, and Deepjyoti **Deka**. "Integrated multi-scale data analytics and machine learning for the distribution grid and building-to-grid interface". In: 2017.

Selected Invited Talks

January 2026 "Powering datacenters: problems and solutions", *IAP Course on Energy Systems Modeling*, MIT.

January 2026 "Risk and impacts of datacenters on grids", *G&T Accounting and Finance Association (virtual)*.

October 2025 "Bayesian neural networks for data-driven opf with severely limited data", *INFORMS*, Atlanta, USA.

April 2025 "Low Carbon Road Transportation: meeting Taiwan's net-zero goals", *Tiger Project Workshop*, Taipei, Taiwan.

Feb 2025 "Optimization Proxies using Limited Labeled Data and Training Time – a Bayesian Neural Network Approach", *AMLD Workshop EPFL*, Lausanne, Switzerland.

Dec 2024 "Realistic Learning in Power Distribution Grids: a physics-informed machine learning perspective", *Analog Devices*, Boston, USA.

Dec 2024 "Parametric Models of Fairness for Decision-Making Problems: from power systems to vehicle routing and prices", *LIDS Climate Tea Talk*, MIT.

Nov 2024 "Fast data-driven Optimization and Risk estimation using limited data in power grids", *Energy & Infrastructure Systems Group (EIMC2) seminar*, MIT.

Sep 2024 "Data-Driven Optimization Using Limited Data", *Autonomous Energy Systems Workshop, NREL*, Golden, USA.

July 2024 "Gaussian Process for Risk estimation in Power Grids", *IEEE PES-GM*, Seattle, USA.

March 2023 "Realistic estimation in power grids", *Energy Systems Innovation Center Seminar, Washington State University(virtual)*.

February 2023 "Tractable Learning in Dynamical Systems excited by Colored non-white inputs", *SIAM CSE*, Amsterdam, Netherlands.

October 2022 "Data-driven Improvements for Stochastic AC-OPF", *INFORMS*, Indianapolis, USA.

January 2022 "Physics Informed Statistical Machine Learning for Distribution Grid Estimation", *IEEE PES Webinar on Advancement of Machine Learning aided Power System State Estimation (virtual)*.

August 2020 "Data-driven stochastic optimization in power grids with safety guarantees", *IEEE PES-GM (virtual)*.

August 2020 "Statistically Optimal Learning of Topology and Line Impedances in Distribution Grids under Correlated Loads", *IEEE PES-GM (virtual)*.

June 2020 "Provable estimation in distribution grids: a physics-informed statistical learning perspective", *IEEE PES Subcommittee on Big Data & Analytics: Virtual Tutorial series*.

Dec 2019 "Learning and Control from Fluctuations in Distribution grids", *Department Seminar, IIT Bombay and IIT Guwahati*, India.

October 2019 "Faulted Line Localization And PMU Placement in Power Systems", *INFORMS*, Seattle, USA.

August 2019 "Data Analytics in Power Grids: Tractable Algorithms", *DSI Workshop*, Livermore, USA.

June 2019 "Estimation and Optimization in Power Grids: a data-driven approach", *INRIA*, Paris, France.

May 2019 "Learning and Control from Fluctuations in Distribution grids: Graphs, Buildings and MDPs", *Control Seminar, University of Minnesota Twin Cities*, Minneapolis, USA.

February 2019 "Efficient Estimation in distribution grids", *Invited session, ISGT*, Washington DC, USA.

April 2019 "Data Analytics in Power Grids: Tractable Algorithms & Path Forward", *NASPI*, San Diego, USA.

Nov 2018 "Joint structure and parameter estimation in power distribution under limited observability", *INFORMS*, Phoenix, USA.

August 2018 "Joint structure and parameter estimation in power distribution under limited observability", *IEEE PES-GM*, Portland, USA.

July 2018 "Learning and Control in Distribution grids", *MOPTA*, Lehigh University, USA.

July 2018 "Learning with end-users in distribution grids", *International Symposium on Mathematical Programming (ISMP)*, Bordeaux, France.

June 2018 "Power Grid Informed & Measurement Based Machine Learning", *International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*, Boise, USA.

March 2018 "Learning and Control from Fluctuations in Grids: Graphs, Buildings and MDPs ", *Lawrence Berkeley National Laboratory*, Berkeley, USA.

Sept. 2017 "Learning in Linear Dynamical Systems", *INRIA*, Paris, France.

August 2017 "Learning the Network of a Linear Dynamical System from Ambient Fluctuations: Application to Power Grid Dynamics", *Automatic Control Laboratory, ETH*, Zurich, CH.

March 2017 "Topology Learning in Power Grids from Ambient Dynamics", *Banff Workshop on Optimization and Inference for Physical Flows on Networks*, Banff, Canada.

January 2017 "Learning and Control in Physical Flow Networks", *INFORMS ICS Conference*, Austin, USA.

January 2017 "Topology Learning in Power Grids from Ambient Fluctuations", *Grid Science Winter School & Conference*, Santa Fe, USA.

Nov. 2016 "Machine Learning in Power Grids", *Australian National University (ANU)*, Canberra, AU.

June 2016 "Machine Learning in Distribution Networks: Estimation and Security", *INFORMS International Conference*, Kona, Hawaii, USA.

May 2016 "Machine Learning Problems in Power Grids", *Pacific Northwest National Laboratory*, Seattle, USA.

April 2016 "Power Grid Big Data Research at LANL", *Smart Grid Workshop, Texas A&M University*, TX, USA.

January 2016 "Learning and Estimation Problems in Radial Distribution Networks", *Physics Informed Machine Learning Conference*, Santa Fe, USA.

Project Leadership

2026 –2028 co-PI, **MIT Energy Initiative Future Energy Systems Center Project on** coordinated and affordable building-grid operation, *Funding: 300k USD*.

2025 –2026 PI, **MIT Energy Initiative Future Energy Systems Center Project on** Flexibility of Data Centers, *Funding: 110k USD*.

2025 –2026 PI, **MIT Energy Initiative Future Energy Systems Center Project on** Understanding Reliability of Co-located Large Loads, *Funding: 150k USD*.

2025 –2026 co-PI, **MIT Energy Initiative Seed Grant on** Leveraging LLMs for Grid Planning, *Funding: 175k USD*.

2025 –2026 co-PI, **MIT GE-Alliance Grant on** AI for Large-Scale, Topology-Aware AC-OPF in Distribution Grids, *Funding: 300k USD*.

2024 –2026 PI, **LANL MIT Grant on** Agentic AI for surrogate model development, *Funding: 360k USD*.

2022 –2025 co-PI, **LANL Laboratory Directed Research & Development Grant on** Machine Learning for Non-convex Optimization, *Funding: 1 million USD*.

2021 – 2024 PI, **LANL Laboratory Directed Research & Development Grant on** Resilient operation of interdependent engineered networks and natural system, *Funding: 1 million USD*.

2020 –2023 LANL co-PI, **US Department of Energy Advanced Grid Modeling Grant on** Machine Learning for Stochastic Optimization, *Funding: 1.3 million USD*.

2020 –2023 LANL co-PI, **US Department of Energy Advanced Grid Modeling Grant on** Real-Time Control of Large-Scale Power Grids, *Funding: 2.5 million USD*.

2018 – 2021 co-PI, **LANL Laboratory Directed Research & Development Grant on** Statistical Machine Learning for Cyber-Physical Systems, *Funding: 900k USD*.

2018 – 2019 co-PI, **US Department of Energy GMLC Grant on** Machine Learning for Distribution Systems, *Funding: 240k USD* .

Student Leadership

2013 – 2015 **Director of Academic Affairs** and **Executive member** of Graduate Student Assembly, The University of Texas at Austin for academic years 2013-2014 and 2014-2015.

2013 – 2014 Member of **President's Student Advisory Council** and **Provost's Student Advisory Council,** The University of Texas at Austin.

2012 – 2013 Member of Graduate School's **Dean's Student Advisory Council** and **Student Senate,** The University of Texas at Austin.

2008 – 2009 Member of **Technical Board of Student Senate,** IIT Guwahati.

Software Skills

Software Matlab, Python, Julia, Gurobi, Ipopt, PowerWorld, Wireshark, TensorFlow

Certification Coursera Deep Learning Specialization

Professional Service

Associate Editor IEEE Transactions on Smart Grid (2021-2022)

TPC Member IEEE Smartgridcomm 2025,2020, 2019, 2018, PMAPS 2020, IEEE GlobalSIP 2019, 2018, 2017, 2016

Organizer Workshop on Machine Learning in Distribution Grids in IEEE Smartgridcomm 2020, Control and Demand Response in Smart Grids session (CDC 2019), Cyber-Physical Resilience in Smart Grids session (Resilience Week 2019), LANL Grid Science Winter School (2017,2019,2021,2025) Machine Learning in Power Grid sessions (ISMP 2018), Energy session (MOPTA 2018) PMU analytics session (INFORMS 2017), LANL Physics Informed Machine Learning Conference 2016

Reviewer **Journal:** IEEE Proceedings, IEEE Access, IEEE Transactions on Power Systems, Transactions on Smart Grid, Transactions on Automatic Control, Transactions on Control of Network Systems, Transactions of Signal Processing, Transactions on Network Science and Engineering, IET Cyber-Physical Systems, IET Generation, Transmission & Distribution, Sustainable Energy-Grids and Networks, Journal of Electrical Power and Energy Systems, Sensors
Conference: IEEE CDC, ACC, CCTA, ECC, PSCC, SmartGridComm, ISIT, PES GM, ISGT, CAMSAP, SECON, GlobalSIP

References

Available upon request